MIL-W-60002A (MU)
4 May 1966
SUPERSEDING
MIL-W-60002 (MU)
13 November 1963

MILITARY SPECIFICATION

WARHEAD , HE , M1 8
PARTS FOR

1. SCOPE

1.1 This specification covers special requirements not normally required by the drawings and provides the quality assurance provisions for the fabrication of parts, assembly anti packing of one type of high explosive warhead designated as Warhead, HE, M18 and is intended for use in the Rocket, HE, 66MM, Anti-Tank, M72.

2. APPICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part" of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2250	Ammunition and Special Weapons; General Specification for.
MIL-D-45608	Designs and Drawings, Ordnance Inspection Equipment.
MIL-I-45607	Inspection Equipment, Supply and Maintenance of.
MIL-P-10025	Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 -	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 -	Quality Assurance Terms and Definitions.
MIL-STD-1167 -	Ammunition Data Cards.

FSC 1340

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MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

10048538 - Body Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL - 10048538 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Parts. - All parts used in the assembly of the warhead shall meet all requirements of applicable drawings and specifications.

3.2 Assembly. - The assembled body shall comply with all requirements specified on drawing (dwg.) 10048538, and with all requirements specified in applicable specifications.

3.3 Leakage. - The body assembly shall be air tight when tested as specified in 4.4.1.

3.4 Workmanship. - All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the

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cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. -Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the. specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product. -At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification

b. Number of units of product inspected.

C* Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a

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guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial Production Inspection

4.2.1 Submission. -At the start of regular production the contractor shall submit 10 complete assemblies, 10 sets of subassemblies (each set of subassemblies shall consist of every subassembly covered by the drawing package) and 10 complete set: of parts to a Government approved facility as designated by the contracting officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of material that has been produced by the contractor or furnished by a supplier and manufactured using the same production process, procedures and equipment that will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply that will be used in regular production. The initial production sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for 'a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply so as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects shall not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed. -Each assembly in the initial production submission shall be examined for every drawing dimension and every drawing and specification requirement. The entire sample shall be examined in accordance with 4.3.2 and then tested as specified in '4.3.3.1.

4.2.3 Rejection of initial production sample. -The initial production sample shall be rejected if any assembly fails to conform to the applicable requirements. The initial production lot shall be rejected if any assembly fails to conform as required when inspected according to 4.3.2 and 4.3.3.1. The Government

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reserves the right to terminate its inspection upon observing any non-conformance of the sample to the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production' sample or portion thereof. Until an initial production sample is accepted, the contractor is in no way authorized by the Government to initiate regular production (including procurement of materials by the contractor) unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 Lot formation. -The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot, Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies , other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

4.3.2 Examination. -Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the Procuring activity. Also, at the option of the procuring

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activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. *Once* agreement has been reached as to which method will be used (group or individual) that method will be used for all of the CD'S listed in this specification. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Liner (See dwg. 10048544 covering a detail of dwg. 10048540).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	**None defined**		
Major:	**AQL 1.00 percent**		
101.	Excessive wall thickness variation	Gage	01001
102.	Length to large datum diameter	Gage	01002
103.	Length to small datum diameter	Gage	01003
104.	Diameter of body, head mating surface, minimum (min)	Gage	01004
105.	Component split, scarred, pitted or dented	Visual	01005
Minor:	**AQL 1.50 percent**		
201.	Location of conduit hole	Gage	01006
202.	Evidence of poor workmanship (See 3.4)	Visual	01007
203.	Wall thickness	Gage	01008

4.3.2.2 Body, head (See dwg. 10048546 covering a detail of dwg. 10048540).

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Categories	Defects	Method of Inspection	Code No.
102.	Wall thickness, min.	Gage	02002
103.	Outside diameter, small end.	Gage	02003
104.	Profile of metal-matic groove incorrect.	Gage	02004
105.	Length to shoulder from rear face.	Gage	02005
106.	Component cracked.	Visual	02006
Minor:	AQL 1.50 percent		
201.	Location of conduit hole.	Gage	02007
202.	Location of metal-matic grooves.	Gage	02008
203.	Protective finish missing, incomplete or improper.	Visual	02009
204.	Evidence of poor workmanship (See 3.4).	Visual	02010

4.3.2.3 Ogive (See dwg. 10048542 covering a detail of dwg. 10048538)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Wall thickness, min.	Gage	03001
102.	Outside diameter of nose cap end.	Gage	03002
103.	Angular location of nose cap detents in relation to hole in forward end.	Gage	03003
104.	Length to nose cap detents.	Gage	03004
105.	Profile of forward end.	Gage	03005
Minor:	AQL 1.50 percent		
201.	Concentricity of nose cap seating surface with body head mating diameter.	Gage	03006
202.	Protective finish missing or incomplete,	Visual	03007
203.	Evidence of poor workmanship (See 3.4).	Visual	03008

4.3.2.4 Head, body assembly (See dwg. 10048540 covering a detail of dwg. 10048538).

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Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Extension of conduit at rear face	Gage	04001
102.	Radius to conduit tab (after folding), min	Gage	04002
Minor:	AQL 1.50 percent		
201.	Improper crimp of liner	Visual	04003
202.	Excessive solder or flux	Visual	04004
203.	Protective finish missing or damaged	Visual	04005
204.	Evidence of poor workmanship (See 3.4)	Visual	04006

4.3.2.5 Body assembly, prior to painting (See dwg. 10048538).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 1.00 percent		
101.	Concentricity of small end of head body assembly with tapered sidewall of head body assembly	Gage	05001
102.	Concentricity of liner with head body assembly	Gage	05002
103.	Concentricity of ogive with head body assembly	Gage	05003
104.	Component insecure or missing	Visual/Manual	05004
105.	Conduit damaged or burred	Visual	05005
106.	Excessive sealant at junction of liner and body	Visual	05006
107.	Insulator connector broken	Visual	05013
Minor:	AQL 1.00 percent		
201.	Length from rear face of body to front face of ogive	Gage	05007
202.	Protective coating incomplete or inadequate	Visual	05008

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4.3: 2.6 Body assembly (see dwg. 10048538).

Categories	Defects	Method of Inspection	Code No.
Critical:	**None defined.**		
Major:	**AQL 0.65 percent**		
101.	Large outside diameter.....	Gage	05009
102.	Evidence of poor workmanship (see 3.4).....	Visual	05010
103.	Paint inadequate or incomplete.....	Visual	05011
Minor:	**None defined.**		

4.3.3 Testing

4.3.3.1 Leakage (see 3.3) - Major defect - Code No. 05012.- This test shall be performed 100 percent. The test shall be performed as specified in 4.4.1 using test equipment specified in 4,3.4. Defective assemblies shall be removed from the lot.

4.3.3,2 Visual examination for marking indicating location of ground pin terminal hole (see dwg. 1,0048538) - Major defect - Code No. 05014. -This inspection shall be performed 100 percent. Defective assemblies 'shall be removed from the lot.

4.3.4 Inspection Equipment. -Index of Inspection Equipment Number IEL-10048538 identifies the inspection equipment required to perform the examinations and tests prescribed in this section The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation. -Inspection Equipment drawings and lists provided and revised in accordance with the requirements of 'the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.3.4.2 Supply and maintenance. - Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment. - The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 Leakage. - The body loading assembly shall be secured to the required test equipment and a pressure differential introduced between the rearward explosive charge cavity and all external surfaces including the forward cavity. The test equipment shall be such that when a rejection test standard is tested by an established pressure level differential over a constant minimum time interval, both of which are maintained throughout testing, a consistent rejection signal indication results. Any body assembly which permits a leakage equal to or in excess of the rejection test standard, shall be classed defective. The rejection test standard shall consist of a body assembly which permits no leakage other than that resulting from a .0082 inch maximum diameter hole located in the apex of the liner. Test equipment sensitivity verification shall be performed at one "hour maximum intervals of production and immediately prior to and at the end of acceptance operations for the day.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - The body assemblies shall be packaged, packed and marked for interplant shipment in accordance with Specification MIL-P-10025.

6, NOTES

6.1 Ordering data. -Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Ammunition data cards shall be prepared for each lot in accordance with MIL-STD-1167.

6.2 Inspection code numbers. -The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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6.3 Inspection equipment. -The con tract shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 inspection equipment list. -Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraph 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes. -The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be desigmated in the IEL by the following codes :

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

6.3.3 Army designs. -Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor nay, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

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6. 3.4 Specification control drawing. - Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the "required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5) . Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment. - Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the' latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 contractor designs. -Contractor designs are designs of inspection equipment' for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in" accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCE may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDCF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 36MM microfilm process. Contractors shall "submit one copy of the final design as a rolled set or flat set. Designs shall 'be submitted for approval as specified in 6.3.7.

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6. 3.7 submission of contractor designs. - Unless otherwise specified on the Inspection Equipment List, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN : SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of initial production sampl . - Instructions as to the location for the evaluation of the initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-NI.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a difinitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications , or other data is not to be regarded by implication or otherwise as in any manner licensing the -holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Preparing activity:
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Project No. 1340-A213

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